

Miles Jennings

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EDUCATION

University of California, Irvine

Irvine, CA

B.S Computer Engineering — Deans Honor List

Sept. 2022 – June 2026

Relevant Coursework: Computer Architecture, Processor Hardware/Software Interfaces, Digital Logic Design, Electronics I & II, Semiconductor Devices, Network Analysis I & II, CMOS Circuit Design, VLSI Design

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

Test Engineer Intern

Buena Park, California

Leach International Corp.

March 2026 – Present

- Diagnosed and repaired **8** malfunctioning test systems within **3** months by analyzing undocumented system architectures and troubleshooting **hardware test fixtures**, reducing operator downtime and eliminating production bottlenecks across the test floor
- Designed and deployed an automated **Python**-based capacitor validation and conditioning system integrating an LCR6002, high-voltage power supply, and safety analyzer, eliminating outsourced testing for 490 components and reducing validation costs by **60%**
- Developed a reusable **LabVIEW** state machine interfacing with a **DAQ's digital I/O** to control a solid-state relay, replacing one-off manual test scripts with a standardized framework for future automated production test systems
- Authored detailed electrical wiring diagrams in **Altium** to re-engineer and restructure key component testing machinery, simplifying **signal routing** and reducing future system troubleshooting time

Control Systems Engineer

Irvine, California

HyperXite – SpaceX Hyperloop

Aug. 2025 – June 2026

- Evaluated the team's **Raspberry Pi 5**-based architecture against an **STM32H753ZI** solution, selecting the STM32 for lower power consumption and native **CAN/ADC** support, eliminating external interface hardware and improving embedded system reliability
- Tested and integrated hardware components for critical pod safety monitoring, designing custom **breadboard test setups** and interfacing sensors with an **STM32 microcontroller** using **C/C++** and STM32CubeIDE
- Designed a custom **control PCB** using **Altium** to integrate embedded control hardware, power distribution, and sensor interfaces, consolidating multiple subsystems onto a single board to simplify system integration and improve maintainability

Technology Intern

Long Beach, California

P2S Engineering

June 2025 – March 2026

- Designed and coordinated **low-voltage infrastructure systems** for commercial facilities, resolving multidisciplinary design conflicts between electrical, mechanical, and architectural teams to improve constructability before construction handoff
- Cut regulatory revision cycles by **11%** by implementing design fixes flagged by supervisors and principal engineers in **3D models**, improving design accuracy and shortening project timelines

PROJECTS

AudioVisor | C, nRF52840, PDM, I2C, Embedded Systems

Sept. 2025 – March 2026

- **Led** a five-person engineering team to develop an embedded wearable capable of localizing nearby sound sources, defining hardware architecture, component selection, and firmware integration from concept through prototype
- Developed real-time firmware on the nRF52840 to sample audio over **PDM** and perform embedded signal processing for directional sound localization
- Integrated peripheral communication and sensor interfaces using **I2C** and optimized embedded signal processing to achieve **84%** accuracy in sound localization during prototype testing

Embedded Building Management System | Python, Raspberry Pi, GPIO, LCD, DHT11, PIR

June 2025

- Designed an embedded building management controller integrating environmental and occupancy sensors to **automate** HVAC, lighting, security, and fire alarm operation without manual intervention
- Implemented interrupt-driven event handling and an LCD-based interface with timestamped logging, enabling real-time **monitoring** and diagnostics of system status

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Verilog, RISC-V, MIPS

Software/Tools: LabVIEW, Revit, AutoCAD, Altium, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA